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REVIEW PAPER

Emerging frontiers in ecosystem and biodiversity monitoring using remote sensing and geographic information systems

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ABSTRACT

The accelerating biodiversity crisis, driven by climate change and intensifying anthropogenic pressures, demands accurate, scalable, and dynamic tools to monitor ecosystem health and biological diversity. Remote sensing and geographic information systems have long been pivotal in observing environmental conditions and measuring biodiversity, nonetheless, the fast-paced development of sensing technologies, analytical approaches, and computational power is greatly transforming their purpose in conservation science. This study provides a comprehensive synthesis of next-generation applications of remote sensing and geographic information systems in biodiversity and ecosystem monitoring. The study aimed to gather recent developments in the use of remote sensing and geographic information systems for biodiversity and ecosystem monitoring, thoroughly evaluate existing methods, recognize enduring challenges, and recommend innovative, technology-driven pathways for improving ecological assessments and conservation planning. A notable transition is taking place from standard land cover mapping towards assessing ecological functions, evaluating habitat quality, and detecting environmental changes in near real-time. Innovative technologies, including hyperspectral imaging, drone-based sensing, radar interferometry, three-dimensional laser scanning, and small satellite constellations, are combined with sophisticated computational methods, featuring machine learning, deep learning, spatiotemporal data fusion, and cloud-based geo-processing. These developments are transforming applications ranging from automated species distribution modelling and ecosystem service mapping to structural-functional landscape phenotyping, habitat connectivity assessment, and predictive early-warning systems for biodiversity loss. The merging of datasets with differing resolutions, timeframes, and sensors is promoting the establishment of broad ecological intelligence, which contributes to adaptive conservation strategies and evidence-based environmental governance. Despite these advances, several challenges remain, including algorithmic bias, the harmonization of heterogeneous datasets, limited direct biodiversity proxies, and disparities in access to emerging technologies. Ethical considerations along with the integration of community-driven monitoring frameworks, are essential for ensuring that technological advancements are in harmony with global sustainability goals. Anticipating the future, the integration of sophisticated sensing technologies, artificial intelligence, and cloud computing platforms presents remarkable opportunities to transform biodiversity monitoring and conservation planning. By enabling predictive, adaptive, and near real-time decision-making, these innovations are reshaping strategies for environmental management and the development of resilient socio-ecological systems in the context of rapid global change.

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INTRODUCTION

Climate change and anthropogenic activities are driving unprecedented alterations in global biodiversity and ecosystem functioning. It is anticipated that by the end of the century, the world will see a temperature increase of 3 to 4 degrees Celsius (°C), leading to changes in species' morphology, behavior, and geographic distribution, which will disrupt ecosystem services essential for human welfare (Shivanna, 2022). These ecological changes are further compounded by land-use transformations, where cropland expansion has increased by over 26 percent (%) in protected areas such as Burkina Faso's Bontioli Nature Reserve, coinciding with significant losses in natural vegetation (Kiribou *et al.*, 2024). A significant relationship exists between the rapid growth of the human population and changes in land cover, which increases the stress on natural ecosystems and complicates efforts to conserve biodiversity. Despite nearly 2,000 scientific articles published between 2000 and 2021 on climate and biodiversity conservation, significant geographic and interdisciplinary knowledge gaps remain, particularly in underrepresented regions and complex socio-ecological systems (Farooqi *et al.*, 2022; Depari, 2024). This deficiency in integration has hindered the ability of natural resource managers to create adaptive, data-centric strategies that are responsive to the fluctuations in biodiversity. There is an urgent need for comprehensive frameworks that bridge ecological, technological, and policy perspectives to enable scalable and predictive biodiversity monitoring. Conventional approaches to biodiversity assessment, including field surveys, camera trapping, and ecological sampling, are fundamental yet face certain limitations. Field surveys relying on visual and aural observations are labor-intensive, time-consuming, and constrained to limited spatial and temporal scales, reducing their effectiveness for capturing complex biodiversity patterns (Bell and Malerba, 2025). Additionally, while camera trapping effectively documents species presence, it is hindered by the high costs associated with deployment, ongoing maintenance requirements, and a bias towards detecting more prominent species, often missing out on elusive or less active taxa (Oliver *et al.*, 2023). Furthermore, their restricted temporal coverage can miss rare events and seasonal behavioral shifts. Similarly, traditional ecological sampling requires expert knowledge, involves significant logistical challenges in remote or rugged terrains, and incurs prohibitive costs when repeated surveys are needed for long-term monitoring (Fergus *et al.*, 2024). These restrictions diminish scalability and may cause biodiversity assessments to be incomplete or biased, thereby hindering effective ecosystem management. Recent technological advancements in remote sensing (RS) and geographic information systems (GIS) offer promising solutions to these challenges. RS platforms, including satellite imagery and drone-based systems, now provide unprecedented capabilities to capture real-time ecosystem dynamics across spatial and temporal scales, supported by diverse spectral, spatial, and temporal resolutions (Lausch *et al.*, 2020; Martínez-López *et al.*, 2021; Marvasti-Zadeh *et al.*, 2024). Integrating these innovations with community-driven monitoring frameworks presents a transformative opportunity to develop scalable, adaptive, and data-rich approaches for biodiversity conservation and sustainable land-use planning in the Anthropocene (Weiskopf *et al.*, 2020). These distinct abilities facilitate thorough and scalable observation of ecosystems across multiple levels. RS has the ability to capture both the structural and functional aspects of ecosystems, delivering vital insights into environmental conditions. The integration of vegetation indices significantly enhances the detection of ecological changes relevant to biodiversity (Yan *et al.*, 2025; Ekaputri *et al.*, 2024; Basyuni *et al.*, 2024). Commonly used indices such as normalized difference vegetation index (NDVI), enhanced vegetation index (EVI), and soil adjusted vegetation index (SAVI) quantify vegetation health and coverage (Melnyk and Brunn, 2025). Furthermore, structural proxies based on light detection and ranging (LiDAR) offer three-dimensional data concerning vegetation and landscape characteristics (Keerthinathan *et al.*, 2025). Tracking these changes over time supports understanding of ecosystem responses to natural and anthropogenic pressures. Remote sensing thus enables broad, continuous, and non-invasive monitoring at scales difficult to achieve with traditional methods (Lawley *et al.*, 2016). This feature is fundamental for well-informed conservation and resource management initiatives. As a result, RS platforms are increasingly valuable for advancing ecological research. The integration of diverse data

types enables a complete perspective on the ecosystem vitality and biodiversity dynamics. Moreover, technological advances in artificial intelligence (AI) and machine learning have revolutionized ecological data analysis (Chen *et al.*, 2024; Pichler and Hartig, 2023; Ryo, 2024). These innovations enable automated processing of vast amounts of environmental data with high accuracy. Cloud-based geospatial platforms, like Google Earth Engine, provide scalable infrastructure for managing and analyzing large spatial datasets (Gorelick *et al.*, 2017). Collectively, these technologies enable species distribution modeling, which forecasts the probable locations of species based on environmental factors. They also enable the detection of disturbances, such as deforestation or habitat degradation, by analyzing changes over time. Phenological tracking, or monitoring seasonal biological events, is made more efficient and precise through these tools (Park and Post, 2022). This quick processing facilitates the prompt identification of ecological changes that could otherwise be overlooked. Consequently, conservation planners can make informed decisions more quickly to protect vulnerable species and ecosystems. Automation reduces manual labor and speeds up data interpretation, increasing overall efficiency. The integration of AI and cloud platforms is transforming biodiversity monitoring into a more dynamic and responsive practice (Bibri *et al.*, 2024). These skills are vital for adjusting conservation initiatives as environmental changes accelerate. Ultimately, these technological advances support more effective and proactive biodiversity management worldwide. In spite of these advancements, there is still limited information on how to successfully merge multi-source RS data, functional trait detection, and socio-ecological indicators into unified biodiversity monitoring frameworks, particularly for complex ecosystems and areas in the global south that are deficient in data. This review study uniquely integrates advances in remote sensing, GIS, and AI-driven analytical frameworks with participatory and cross-disciplinary approaches, providing actionable pathways for predictive, high-resolution, and socially inclusive biodiversity monitoring that extend beyond summarizing existing methods. This study provides an extensive and future-oriented examination of emerging RS and GIS technologies, analytical approaches, and applied frameworks for monitoring biodiversity and ecosystem changes. It emphasizes the need to overcome current challenges and to define future research paths that support adaptive, inclusive, and intelligent ecological decision-making systems. This study aims to consolidate recent advancements in RS and GIS applications for the monitoring of biodiversity and ecosystems, critically evaluate existing methodologies, identify persisting challenges, and propose innovative, technology-driven strategies for enhancing ecological assessments and conservation planning. This study has been carried out globally, integrating research findings from diverse ecosystems and regions, and was conducted in the year 2025 to reflect the most recent developments and emerging trends in the field.

METHODOLOGY

Search strategy and sources

A systematic literature search was conducted using four primary academic databases: Scopus, Web of Science, and ScienceDirect. The selection of these databases is based on their broad coverage of environmental sciences, geospatial technologies, and studies related to biodiversity monitoring. To complement these, Google Scholar was used to capture additional grey literature, including technical reports, case studies, and regionally focused investigations not indexed in the primary databases. The strategy for searching included the integration of various keywords along with Boolean operators to enhance the retrieval of pertinent studies. Search terms included combinations such as “remote sensing,” “satellite imagery,” “hyperspectral imaging,” “drone-based sensing,” “biodiversity monitoring,” “ecosystem health,” “species distribution modeling,” “GIS,” “spatial analysis,” “functional ecological assessment,” “real-time change detection,” and “landscape phenotyping.” To include the most recent technological breakthroughs, the search focused on studies released between 2015 and 2025, a span during which there were significant advancements in hyperspectral imaging, drone-based platforms, radar interferometry, Cube Satellite (CubeSat) constellations, and advanced analytical techniques, including machine learning, cloud computing, and spatiotemporal data fusion.

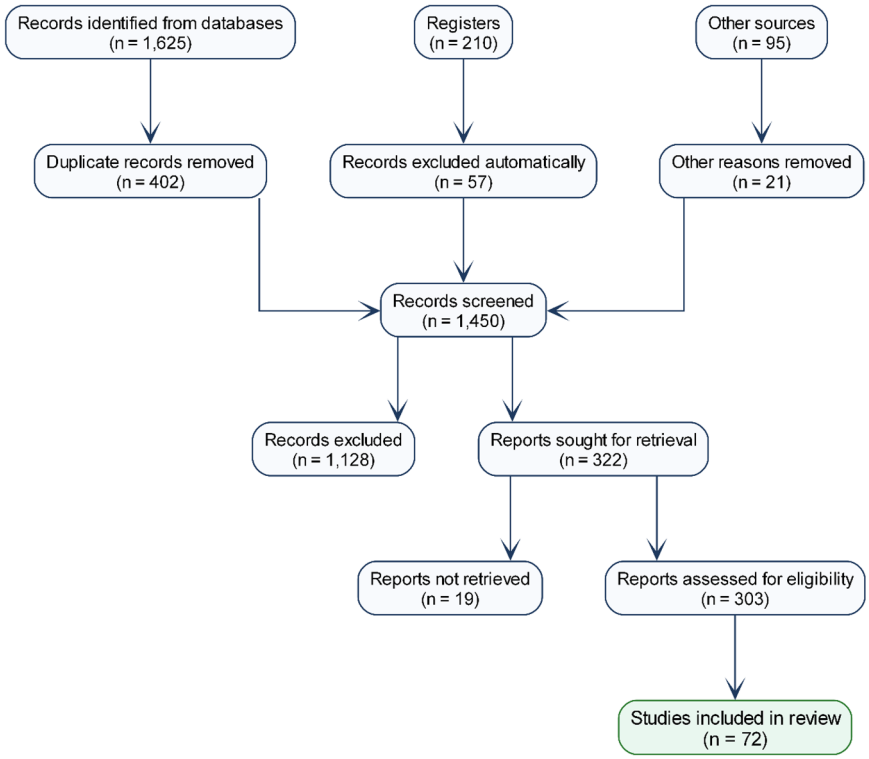

Fig. 1: PRISMA 2020 flow diagram showing the identification, screening, eligibility assessment, and inclusion of studies in the systematic review

Inclusion and exclusion criteria

Studies were accepted if they employed advanced RS or GIS technologies for the purposes of biodiversity monitoring, ecosystem evaluation, or species distribution modeling. Priority was given to studies that demonstrated integration of multi-resolution, multi-sensor, or multi-temporal datasets and those that employed emerging analytical frameworks such as machine learning, deep learning, or cloud-based geospatial analytics. Studies based on empirical evidence, whether they yield quantitative or qualitative outcomes, and systematic reviews that summarize technological innovations, were also viewed as relevant. Studies were excluded if they focused exclusively on traditional land cover mapping without linking results to biodiversity or ecological parameters, lacked methodological clarity, or failed to provide sufficient detail on data sources and analytical approaches. Publications presenting purely theoretical frameworks without empirical validation were also excluded. Moreover, only studies that were published in English were included to guarantee consistency in interpretation, considering the subject's technical complexity. The study selection process for the systematic review is summarized in the PRISMA flow diagram (Fig. 1), illustrating the numbers of records identified, screened, assessed for eligibility, and ultimately included.

Study selection, scope, and data synthesis

All search results were transferred to Zotero for organization and systematic screening, with duplicates cleared and the remaining studies initially reviewed by title and abstract for relevance. Full-text articles were then evaluated against predefined inclusion criteria, prioritizing studies employing advanced technologies such as hyperspectral imaging, drone-based sensing, radar interferometry, and CubeSat constellations, particularly those demonstrating novel applications in biodiversity monitoring, species distribution modeling, structural-functional landscape phenotyping, or

real-time ecological change detection. The review covered a variety of ecosystems and geographical areas, such as biodiversity hotspots, protected regions, urban settings, and marine environments, focusing on research from low- and middle-income countries where limited data and infrastructure issues hinder extensive monitoring. For each study, detailed information was extracted on the type of RS or GIS technology, dataset resolutions, analytical frameworks, biodiversity metrics, sampling protocols, objectives, key findings, limitations, and potential policy implications. Due to the methodological diversity, a qualitative synthesis was performed to uncover shared themes, emerging technological trends, analytical progress, and ongoing knowledge deficiencies.

Technological convergence in ecosystem and biodiversity monitoring

Hyperspectral and multispectral satellite imaging

Recent progress in hyperspectral and multispectral satellite imaging has notably broadened the scope of remote sensing in ecological monitoring, shifting from standard broadband indices to the classification of species and biochemical characteristics. For instance, the use of spaceborne sensors such as precursore iperspettrale della missione applicativa (PRISMA), DLR earth sensing imaging spectrometer (DESIS), and environmental mapping and analysis program (EnMAP) has demonstrated exceptional potential for identifying vegetation characteristics, detecting plant physiological traits, and delineating individual species. Lassalle *et al.* (2023) illustrated this by achieving classification accuracies of up to 97% (OA) and 0.95 (Kappa) in mangrove species mapping using multisource data, including DESIS and PRISMA. Their findings also highlight the superior performance of hyperspectral imagery at sub-metric resolutions with 0.5 meters (m), especially when incorporating short-wave infrared (SWIR) bands. Furthermore, Lu *et al*. (2020) pointed out the importance of hyperspectral imaging in agriculture for identifying minor spectral variations related to crop biochemical and morphological features, facilitated by the rise of compact, cost-effective airborne sensors and improved access to satellite data. The EnMAP mission further reinforces this technological leap, offering 30 m spatial resolution and high spectral sampling with 6.5–10 nanometers (nm) across the VNIR and SWIR regions (Chabrillat *et al.*, 2024). Since its operational launch in 2022, EnMAP has provided high signal-to-noise, globally accessible data that support diverse environmental applications, including vegetation analysis, soil composition mapping, and water quality monitoring. Fig. 2 classifies the new remote sensing and GIS technologies, analytical methods, applications, and challenges, underlining their interconnected roles in the observation of biodiversity and ecosystems.

Radar and LiDAR for structural complexity

Forest structural complexity serves as a vital determinant of both biodiversity and ecosystem performance, however, effectively measuring this complexity on a large scale continues to be a challenge. Synthetic aperture radar (SAR) and light detection and ranging (LiDAR) technologies have emerged as powerful tools to characterize forest structure, above-ground biomass, and habitat heterogeneity. In particular, LiDAR supplies exact three-dimensional information that supports the estimation of vital forest features such as canopy height, biomass, and volume, which are deeply connected to habitat quality. Coops *et al.* (2021), demonstrate that LiDAR-derived models can explain a substantial portion of the variability in forest height (mean R^2 = 0.64) and biomass (mean R^2 = 0.55) across diverse forest landscapes, highlighting the technology's capacity to generate spatially explicit, accurate structural metrics. Achieving this level of precision is vital for grasping the complexity of habitats in both managed and natural woodlands. The combination of LiDAR and optical remote sensing significantly improves the capacity to observe structural dynamics across extended time periods. Tanase *et al.* (2024), effectively combined airborne and space borne LiDAR with Landsat time series to reveal significant increases in forest canopy cover and biomass over nearly four decades in the Iberian Peninsula. Their findings, including an 11% increase in canopy cover and biomass rising from approximately 32 to 50 tons per hectare (t/ha), underscore the power of combining spectral and structural data to track forest recovery and growth trends relevant for biodiversity assessments and carbon accounting. Bae *et al.* (2019), illustrate that habitat structure metrics obtained from radar are comparable to those from airborne LiDAR in forecasting biodiversity patterns across various taxa.

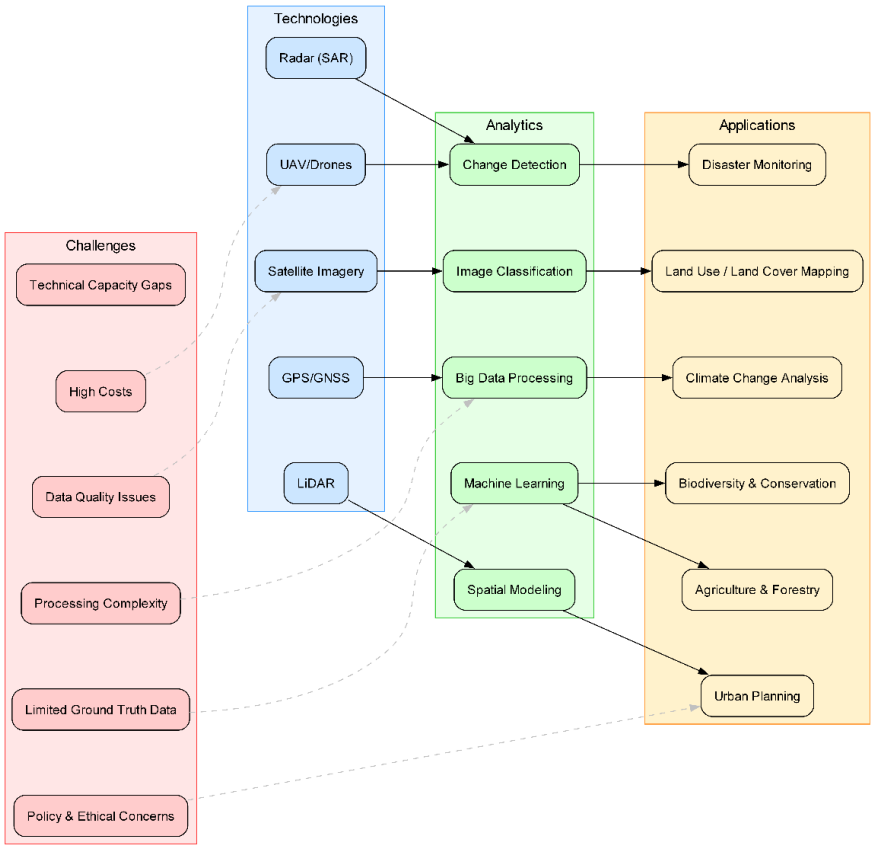

Fig. 2: Key remote sensing and GIS technologies, analytics, applications, and challenges

Their results indicate that SAR achieves comparable, and in some cases superior, predictive power (median R^2 = 0.57 vs. 0.51 for LiDAR) for species assemblage composition, emphasizing radar's suitability for large-scale biodiversity monitoring where cloud cover or logistical constraints limit LiDAR acquisition. Radar's capability to penetrate canopies and react to the moisture and structure of vegetation offers distinct, supplementary insights that improve ecosystem characterization. Advances in multi-sensor fusion further leverage the strengths of these technologies. Lutz *et al.* (2024), show that integrating global ecosystem dynamics investigation (GEDI) LiDAR canopy height data with sentinel-1 synthetic aperture radar (SAR) and multispectral imagery can robustly estimate aboveground biomass density across varied vegetation types in Western Australia, ranging from tall eucalypt forests to savanna grasslands.

UAVs and CubeSat constellations

The introduction of cost-effective unmanned aerial vehicles (UAVs) and high-frequency CubeSat constellations including PlanetScope has changed the landscape of environmental monitoring on both local and regional scales, offering exceptional spatial and temporal resolution. These technologies enable rapid, cost-effective surveillance that is critical for detecting subtle and transient ecological phenomena, disturbances, and illegal activities, which are often missed by traditional satellite systems with coarser revisit times or spatial footprints. UAVs that are equipped with multispectral sensors enable highly detailed observations at the required scales for precision agriculture and habitat monitoring. In contrast, CubeSat constellations deliver daily global coverage with moderate spatial resolution, which supports near real-time evaluation of phenological

and land cover dynamics. Moletto-Lobos *et al.* (2024), highlight the complementary capabilities of UAV multispectral data and PlanetScope satellite imagery in monitoring winter wheat phenology and fertilization practices in Mediterranean agroecosystems. Their findings demonstrate that PlanetScope's moderate resolution (approximately 3-5 m) is sufficient for capturing canopy cover variations (R^2 = 0.68) and crop yield estimates (R^2 = 0.54) across heterogeneous fields, although it tends to smooth over fine-scale treatment variability that UAVs can better resolve. This complementarity underscores the potential for integrated UAV and CubeSat data streams to optimize agricultural management and enhance climate-resilient practices through continuous monitoring of crop status and productivity. Furthermore, CubeSat constellations have shown to be valuable for monitoring biodiversity-related phenology beyond agricultural uses. Miura *et al.* (2023), effectively utilized PlanetScope's imagery, which offers high temporal and spatial resolution, to identify general flowering events in the dipterocarp forests of Borneo. They were able to capture both the timing and spatial extent of synchronous mass flowering, revealing moderate to strong correlations (0.52–0.85) to in situ phenology observations. In semi-arid environments, where phenological patterns are spatially heterogeneous and temporally irregular, Liu *et al.* (2024), validate PlanetScope's ability to detect plant-specific phenology metrics, such as green up and senescence timings, with good agreement to ground-based PhenoCam observations (average R^2 up to 0.84 during green up phases). Even with ongoing delays in recognizing the onset of senescence and dormancy, PlanetScope's dependable performance and precise temporal resolution allow for intricate monitoring of vegetation dynamics in challenging ecosystems, which is essential for adaptive approaches in land and water management. Fig. 3 presents an integrated RS–GIS framework for biodiversity monitoring and conservation.

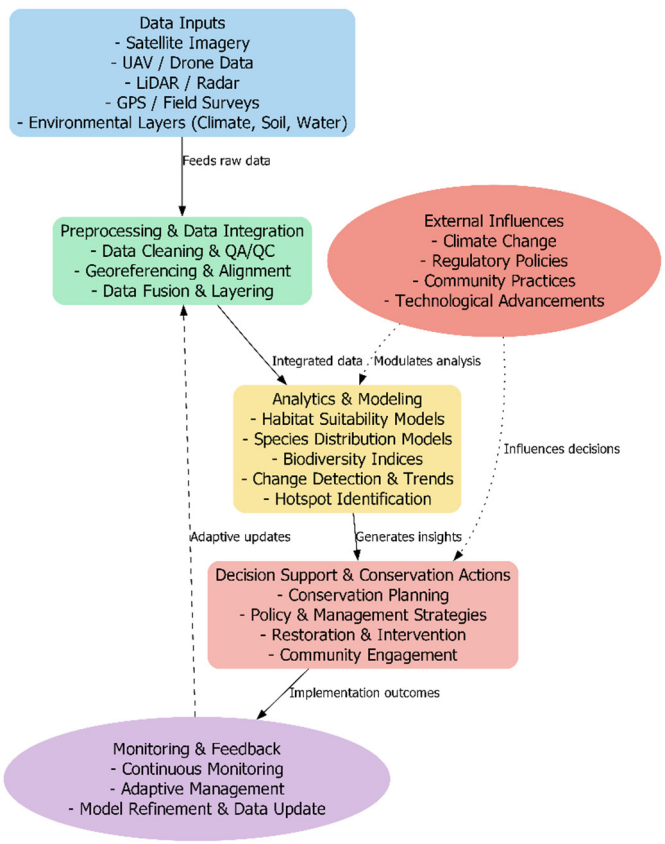

Fig. 3: Integrated RS–GIS framework for biodiversity monitoring and conservation

Moving beyond NDVI

Despite its importance, the NDVI has limitations in assessing complex physiological states such as canopy water content, stress responses, and photosynthetic efficiency, even though it is a key tool in vegetation monitoring due to its simplicity and robustness. Consequently, more refined vegetation indices like the EVI, photochemical reflectance index (PRI), normalized difference water index (NDWI), and red-edge based metrics have been increasingly adopted to provide nuanced insights into canopy stress, water status, and productivity beyond what NDVI can reliably detect. Zhang and Zhou (2019), indicated that indices focused on vegetation water content, especially those employing red-edge spectral bands such as the green chlorophyll index (CIgreen), red edge normalized ratio (NRred edge), and red-edge chlorophyll Index (CIred edge), exhibited greater sensitivity to fluctuations in canopy water content (CWC) under water stress conditions in summer maize. It was found that CWC presented a significant single-peak seasonal fluctuation and was the most reliable predictor of crop water status among the investigated parameters. Their study emphasized the optimal predictive power of these indices over others like leaf equivalent water thickness (EWT) and live fuel moisture content (LFMC). In a related study, Zhu *et al*. (2024) pointed out the effectiveness of indices that utilize near-infrared (NIR) and shortwave infrared (SWIR) bands, including the Normalized Difference Infrared Index (NDII), Moisture Stress Index (MSI), Photochemical Reflectance Index (PRI), Land Surface Water Index (LSWI), and Global Vegetation Moisture Index (GVMI) in the context of for drought monitoring in grasslands. Their coupled radiative transfer modeling and satellite band evaluation showed silhouette coefficients exceeding 0.7 for NDII, MSI, PRI, LSWI, and GVMI during the stable growth stage, indicating high sensitivity to drought conditions. Among satellites analyzed, MODIS Band 7 showed the highest sensitivity to water content, which further emphasizes the importance of these indices for drought detection over standard NDVI. Further supporting this, Zhou *et al.* (2022), conducted irrigation experiments on maize and demonstrated that while NDVI and other traditional indices exhibited some sensitivity to canopy water content, the Water Index (WI) and NDWI provided stronger linear relationships with CWC, with coefficients of determination (R^2) around 0.55 to 0.57 within a CWC range of 0.01-0.41 kilogram per square meter (kg/m²). Conversely, both NDVI and the Optimal Soil-Adjusted Vegetation Index (OSAVI) demonstrated similar but diminished predictive strength, with key threshold values identified at 0.55 kg/m² for vegetation water content. These indices also reflected dynamic water content changes better than NDVI throughout the growing season, underscoring their utility in agricultural drought assessment and irrigation management.

Machine learning and deep learning applications

Machine learning (ML) and deep learning (DL) play a crucial role in environmental and ecological studies, providing robust tools for classification, anomaly detection, and predictive modeling with impressive precision. Talaei Khoei and Kaabouch (2023), compared a suite of supervised and unsupervised models on the Canadian Institute for Cybersecurity Distributed Denial of Service 2019 Dataset (CICDDOS 2019) benchmark dataset for intrusion detection, demonstrating that the Alex Neural Network (a deep learning model) achieved the highest detection accuracy among supervised models, outperforming traditional algorithms such as Gaussian Naïve Bayes and Support Vector Machines (SVM). Regarding unsupervised learning, the Variational Autoencoder model exhibited outstanding performance, highlighting the enhanced capability of deep architectures in identifying intricate patterns without relying heavily on labeled data. This highlights the potential of DL for ecological anomaly detection where labeled datasets are often scarce. This study, although centered on cybersecurity, illustrates the ability of deep learning architectures for detecting complex patterns in high-dimensional datasets. By analogy, similar approaches could be applied to biodiversity monitoring, where ecological anomaly detection often faces scarce labeled data and complex, nonlinear relationships among environmental variables and species observations. Chowdhury *et al*. (2024) pointed out the significance of merging machine learning and artificial intelligence with geographic information systems and remote sensing in their examination of coral reef surveillance, particularly to track the health of coral reefs facing the pressures of climate change. The study identified Random Forests, Convolutional Neural

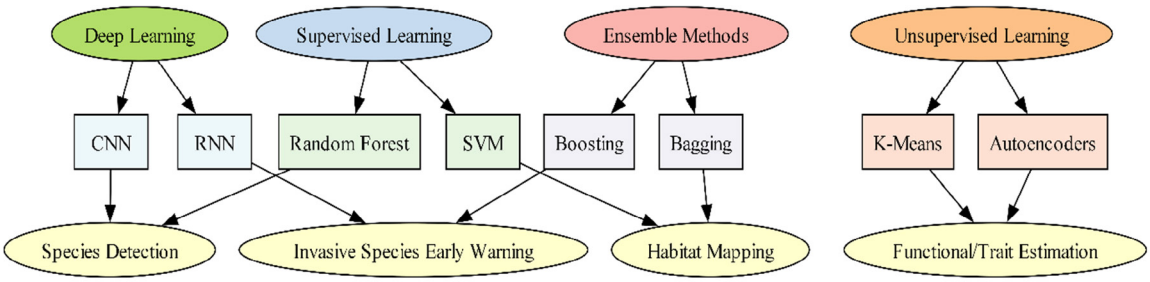

Fig. 4: Conceptual overview of innovative machine learning approaches in RS–GIS for biodiversity monitoring– Recurrent neural network (RNN)

Networks (CNNs), and transfer learning as dominant frameworks enabling high-accuracy classification of reef conditions, detection of bleaching events, and habitat degradation. Fig. 4 presents a conceptual framework of advanced machine learning strategies implemented in RS–GIS for biodiversity monitoring, demonstrating the links between categories of algorithms, specific methodologies, and their key ecological applications.

Table 1 provides an overview of innovative machine learning approaches integrated with RS–GIS technologies for enhancing biodiversity monitoring and assessment.

Spatiotemporal data fusion

Techniques for spatiotemporal data fusion, including STARFM and BFAST, have substantially increased the capability for ongoing landscape monitoring of landscapes and the early detection of environmental changes by skillfully combining coarse-resolution and fine-resolution remote sensing imagery. Wu *et al.* (2020), applied the breaks for additive seasonal and trend (BFAST) algorithm integrated with random forests to monthly Landsat NDVI time series over the West Dongting Lake wetland in China, achieving an overall accuracy of 87.8% in detecting multiple types of forest changes including deforestation, reforestation, and flooding-related variations between 2000 and 2018. The study's outcomes demonstrated clear spatiotemporal forest dynamics, featuring rapid plantation growth from 2000 to 2005, a stabilization phase from 2006 to 2011, and a decline mainly caused by human-induced factors. Building on the need for efficient and scalable change detection, Masiliūnas *et al.* (2021), introduced BFAST Lite, a lightweight unsupervised time series analysis method optimized for speed and flexibility. Tested against an 11-year MODIS dataset with over 30,000 global land cover change points, BFAST Lite matched the accuracy of the original BFAST but significantly outperformed it in processing speed and robustness to missing data. This algorithm reached a perfect balance between user and producer accuracy by using alternative alternative information criteria to identify change breaks, making it particularly effective for large-scale environmental monitoring. More recently, Li and Wu (2024), developed Multi-RLD-Net, a deep learning-based multi-task network for precise change area identification and exact change moment detection in multispectral time series imagery from platforms such as Landsat and Sentinel-2. By combining Optical Flow and Long Short-Term Memory (LSTM) networks, Multi-RLD-Net effectively aligns temporal features and captures long-term dependencies, resulting in a substantial performance gain. Their experiments conducted on benchmark datasets (UTRNet and DynamicEarthNet) revealed enhancements in in F1 scores of 1.29% and 10.42% compared to previous leading method MC2ABNet, respectively, emphasizing the strength of sophisticated spatiotemporal fusion integrated with deep learning for comprehensive environmental change analysis (Fig. 5).

Applications in biodiversity and ecosystem monitoring

Species distribution and habitat risk mapping

RS-GIS tools support dynamic modeling of species distributions under climate and land-use change scenarios, aiding conservation prioritization. These tools facilitate the incorporation of various environmental factors and spatial information to forecast possible changes in the suitable habitats

Table 1: Innovative machine learning approaches in RS–GIS for biodiversity

Methodology	Description	Data requirements	Typical algorithms / models	Advantages	Limitations	Applications in Biodiversity	Sources
Transfer Learning	Reusing models trained in data-rich regions or large datasets for data-poor biodiversity regions.	Labeled RS–GIS data from source domain; minimal local labeled data.	Pretrained CNNs (Residual Network, Visual Geometry Group Network), Vision Transformers.	Reduces local data needs; accelerates deployment.	Domain mismatch risk; requires adaptation.	Mapping species in under-surveyed regions; habitat classification.	Kaya *et al.*, 2019; Sun *et al.*, 2022; Tariku *et al.*, 2023
Explainable AI (XAI)	Methods that reveal why models make certain predictions.	Labeled imagery; model predictions and features.	SHapley Additive exPlanations, Local Interpretable Model-Agnostic Explanations, Gradient-weighted Class Activation Mapping, Integrated Gradients.	Builds trust; identifies biases; supports regulation.	May lower performance; explanations may be complex.	Policy-compliant habitat mapping; identifying bias in endangered species models.	Ali *et al.*, 2023; Haque *et al.*, 2023; Saeed and Omlin, 2023
Ensemble Learning	Combining multiple models to improve predictive accuracy.	Any labeled biodiversity RS–GIS dataset.	Random Forests, Gradient Boosting, Model Averaging, Stacking.	Reduces overfitting; more robust predictions.	Higher computation; less interpretable.	Forest type mapping; multi-species habitat prediction.	Ganaie *et al.*, 2022; Mienye and Sun, 2022; Mohammed and Kora, 2023
Unsupervised Learning for novelty detection	Detecting rare or unexpected patterns without prior labels.	Unlabeled RS–GIS or sensor data.	Density-Based Spatial Clustering of Applications with Noise, Isolation Forest, k-Means, Self-Organizing Maps.	Finds new species/events; early warnings.	Difficult validation; false positives.	Rare species detection; invasive species alerts.	Bergmann *et al.*, 2021; Eltouny *et al.*, 2023; Hossen *et al.*, 2024
Data augmentation and synthetic data generation	Expanding datasets via transformations or synthetic generation.	Limited labeled data.	Generative Adversarial Networks, StyleGAN, image rotation/scaling methods.	Overcomes scarcity; boosts robustness.	Synthetic data may lack realism.	Rare species detection in low-data regions.	ANNAKI *et al.*, 2024; Sung *et al.*, 2022
Few-shot learning	Learning from a few examples per class.	Few labeled examples, plus background data.	Matching Networks, Prototypical Networks.	Useful for rare/endangered species.	Sensitive to variability.	Identifying rare orchids from aerial imagery.	Song *et al.*, 2023; Tian *et al.*, 2024; Ye *et al.*, 2024
Semi-Supervised Learning	Leveraging small labeled + large unlabeled datasets.	Large unlabeled imagery, small labeled subset.	Semi-supervised CNNs, Mean Teacher, Pseudo-labeling.	Reduces annotation costs; boosts accuracy.	Unlabeled data quality critical.	Mapping understudied habitats.	van Engelen and Hoos, 2020; Verma *et al.*, 2022; Yang *et al.*, 2023
Self-Supervised Learning	Pretraining with proxy tasks, then fine-tuning.	Large unlabeled datasets.	Simple Framework for Contrastive Learning of Visual Representations, Momentum Contrast, Bootstrap Your Own Latent.	Avoids costly labeling; uses massive RS data.	Requires relevant pretext tasks.	Broad-scale vegetation mapping.	Jaiswal *et al.*, 2020; Liu *et al.*, 2021; Rani *et al.*, 2023
Active Learning	Selecting most informative samples for labeling.	Initial small labeled dataset + large unlabeled pool.	Uncertainty Sampling, Query-by-Committee.	Minimizes labeling effort; maximizes gain.	Needs expert feedback loop.	Targeted species image annotation.	Mendonça *et al.*, 2023; Ren *et al.*, 2022
Federated Learning	Training across distributed datasets without sharing data.	Local datasets at different institutions.	Federated Averaging, Federated Proximal.	Data privacy; collaborative modeling.	Model drift risk; high communication cost.	Cross-country biodiversity monitoring.	Li *et al.*, 2023; Yang *et al.*, 2019; Zhang *et al.*, 2021

Table 1: Innovative machine learning approaches in RS–GIS for biodiversity

Methodology	Description	Data requirements	Typical algorithms / models	Advantages	Limitations	Applications in Biodiversity	Sources
Graph Neural Networks (GNNs)	Modeling spatial and ecological relationships as graphs.	Species occurrence + environmental/spatial data.	Graph Convolutional Networks, Graph Sample and Aggregate, Graph Attention Networks.	Captures species interactions.	Graph construction complexity.	Migration corridor prediction.	Wu *et al.*, 2023, 2021; Zhou *et al.*, 2020
Spatio-Temporal Deep Learning	Models that integrate both spatial and temporal biodiversity patterns.	Multi-timepoint imagery, environmental data.	Convolutional Long Short-Term Memory, Temporal CNN, Transformers.	Captures seasonal/migratory changes.	Computationally heavy.	Seasonal wetland monitoring.	Amato *et al.*, 2020; Wang *et al.*, 2022; Zhang *et al.*, 2020
Hybrid Physics–ML Models	Combining ecological process models with ML predictions.	RS–GIS + ecological simulation outputs.	ML + species distribution models.	Adds interpretability; respects ecology.	Integration complexity.	Climate change impact assessment.	Arcomano *et al.*, 2022; Rai and Sahu, 2020; Willard *et al.*, 2023
Domain Adaptation with Adversarial Training	Aligning feature spaces across domains.	Source + target domain data.	Domain-Adversarial Neural Networks.	Improves generalization.	Training instability.	Sensor-to-sensor habitat mapping.	Fujii *et al.*, 2022; Liang *et al.*, 2023
Reinforcement Learning for Conservation	Optimizing monitoring and conservation strategies.	Historical action-outcome data.	Deep Q-Networks, Policy Gradient.	Adapts to dynamic changes.	Needs accurate simulation.	Patrol route optimization for poaching prevention.	Equihua *et al.*, 2024; Lapeyrolerie *et al.*, 2022
Anomaly Detection with Autoencoders	Identifying deviations from normal patterns.	Historical baseline imagery.	Variational Autoencoders, Deep Autoencoders.	Good for disturbance alerts.	Noise sensitivity.	Detecting forest fires or illegal logging.	Tien *et al.*, 2021; Tziolas *et al.*, 2022
Multi-Modal Learning	Integrating multiple data sources.	Optical, LiDAR, SAR, citizen science.	Fusion CNNs, Transformer fusion models.	Richer, more robust predictions.	Data alignment complexity.	Integrated habitat classification.	Audebert *et al.*, 2018; Wang *et al.*, 2023; Wen *et al.*, 2024
Uncertainty Quantification	Measuring confidence in predictions.	Model predictions + probability outputs.	Bayesian Neural Networks, Monte Carlo Dropout.	Prioritizes reliable outputs.	Can be misleading in noisy data.	Targeting high-certainty species detection for surveys.	Abdullah *et al.*, 2024; Fiedler and Lucia, 2023; Hirsh *et al.*, 2022
Edge AI for Field Monitoring	Running models on field devices.	Compressed model versions + sensor data.	TensorFlow Lite, TinyML.	Enables real-time detection.	Hardware limits.	Drone-based wildlife tracking.	Arashpour, 2023; Kim *et al.*, 2023

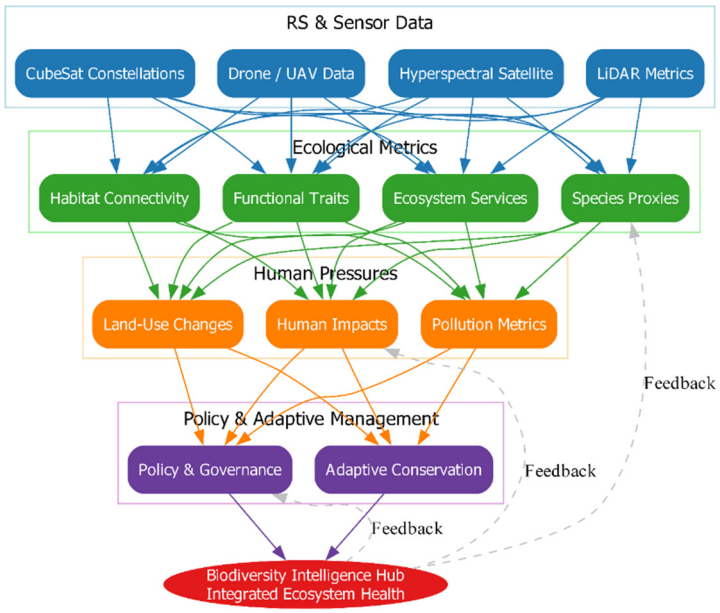

Fig. 5: Multi-dimensional biodiversity intelligence network

for species over time, which is essential for efficient forest management and the preservation of biodiversity. Supporting this approach, Booth (2018), demonstrates that species distribution models (SDMs) have advanced significantly, with correlative SDMs now commonly applied to analyze climate change impacts on more than 50,000 tree species globally. Over 1,000 tree species have been evaluated for anticipated shifts in suitable habitats, demonstrating the increasing ability of SDMs to assist in forest management in light of future climate scenarios. Furthermore, while SDMs are often scrutinized, the limitations identified tend to be less pronounced than commonly asserted, especially when species occurrence data is combined with climatic, soil, and provenance trial data. Importantly, the integration of environmental and species data enhances the accuracy of habitat suitability mapping, facilitating targeted conservation efforts. Complementing this, Anselmetto *et al.* (2025), highlight the value of local-scale species data in improving model performance for current and future species distributions. Their research involving 22 tree species in the Alpine area of Piedmont indicates that SDMs developed using detailed local forest inventories surpassed those based on coarse-scale models, particularly in forecasting future alterations under the RCP4.5 and RCP8.5 climate scenarios. However, they note that models trained on local data may suffer from niche truncation when extrapolating beyond current conditions, which can lead to underestimation or overestimation of range shifts. In a wider biogeographical framework, Franklin (2023) examines three decades of SDM applications and emphasizes their crucial importance in comprehending species distributions across past, present, and future contexts. SDMs are instrumental for delineating ecoregions, assessing biodiversity conservation status, and forecasting species range shifts due to global changes. Issues like defining niches, ensuring data quality, transferring models, and selecting predictors have been increasingly tackled, enhancing the reliability of SDMs for conservation planning. At a landscape scale, García-Amorena *et al.* (2021), combine remote sensing and species distribution modeling to assess the impact of climate change and land use on Pinus hartwegii in Mexico's Izta-Popo National Park. Using 24 Maxent models and Landsat imagery from 1993 and 2013, they reveal significant habitat loss and fragmentation, reducing connectivity and dispersal potential for the species. Projections for 2050 and 2070 under multiple carbon dioxide (CO_2) concentration pathways indicate further

reductions in habitat suitability, with populations likely needing to migrate upslope. Nonetheless, the findings reveal that changes in land use driven by human activity in the past 20 years have concealed some effects of climate change, underscoring the complicated interaction between natural processes and human influences on habitat vulnerability. Table 2 highlights emerging applications of remote sensing (RS) and GIS technologies in advancing biodiversity conservation and ecosystem management.

Monitoring restoration and landscape connectivity

Post-restoration monitoring that incorporates drone-derived data, spectral indices, and LiDAR facilitates the quantification of ecological recovery and functional connectivity. These advanced remote sensing technologies provide high-resolution, scalable, and multidimensional insights into vegetation structure, health, and landscape fragmentation, critical for evaluating restoration success and guiding adaptive management. Robinson *et al.* (2022), emphasize the expanding role of drones in restoration ecology, highlighting their capacity for mapping habitats, tracking restoration progress, and managing disturbances at multiple scales. Considering the anticipated human-induced deterioration of as much as 95% of global ecosystems by 2050, drone technology presents a viable and scalable solution for improving restoration monitoring across the entire restoration process, from planning to execution and continuous evaluation. Their review underscores how drones can significantly improve data collection efficiency and accuracy, essential for meeting ambitious targets such as those set by the United Nations Decade on Ecosystem Restoration. Building on this, Iheaturu *et al.* (2024), demonstrate the power of integrating UAV LiDAR and multispectral data for assessing forest status and disturbance severity in a West African forest patch. By employing principal component analysis, they formulated an integrated disturbance index (IDI) that exceeded the performance of single-sensor methods, reaching a 95% accuracy rate in detecting disturbances. Their findings reveal that nearly 77% of the 560-ha Ewe-Adakplame relict forest faces medium to high disturbance, underscoring the urgency for targeted conservation efforts. Integrating structural LiDAR data with spectral vegetation indices creates a strong foundation for measuring restoration progress and prioritizing interventions for connectivity interventions. Shi *et al.* (2024), further validate the use of combined multispectral remote sensing and laser point cloud monitoring in evaluating vegetation ecological restoration. Applied in China's Daqing oilfield, their approach achieved over 90% monitoring accuracy, revealing a significant vegetation coverage increase of 7,509 km² attributed to farmland conversion to forest.

Urban biodiversity and green infrastructure

Remote sensing and geographic information system applications are gaining importance in urban areas, where they facilitate the planning and implementation of green infrastructure to promote biodiversity, alleviate urban heat, and strengthen ecological connectivity. Czyża and Kowalczyk (2024), demonstrated the use of high-resolution vector data (1 m precision) combined with GIS to identify suitable locations for blue-green infrastructure (BGI) components, including bioswales, infiltration trenches, and green bus stops in Olsztyn, Poland. Their integrative approach combined spatial datasets with a surface urban heat island (SUHI) model and demographic age structure, allowing the identification of priority sites that not only reduce urban heat but also address the social needs of vulnerable populations. This precise spatial modeling aids in the focused deployment of multifunctional green infrastructure, which bolsters ecosystem resilience and enhances the quality of urban life in fast-expanding metropolitan areas. Such comprehensive spatial analyses ensure that blue-green infrastructure is placed effectively to maximize ecological and social benefits in cities. Codemo *et al.* (2022), focused on accurately mapping pervious surfaces and canopy cover to support urban green infrastructure planning through a novel combination of normalized difference vegetation index (NDVI) from airborne imagery at 0.20 m resolution and digital elevation models. Utilized in the Municipality of Trento, Italy, this technique reached an accuracy level greater than 80%, facilitating precise assessment of green space distribution across urban blocks. The resulting maps offer planners detailed insights into areas with deficient canopy or permeable surfaces, essential for guiding interventions that enhance urban climate adaptation and storm water management

Table 2: Emerging applications of remote sensing and GIS in biodiversity conservation

Application area	Description	Potential impact	Key technologies/methods	Current limitations	Examples/use cases	Sources
Climate Change Impact Modeling	Using RS data to model species range shifts, habitat loss, and ecosystem resilience under future scenarios.	Supports anticipatory conservation planning and ecosystem restoration	Satellite climate datasets, species distribution models (SDMs)	Requires integration with complex climate models; uncertainty remains high	Modeling range shifts of forest species under warming	Dubayah *et al.*, 2020; Requena-Mullor *et al.*, 2017; Venter *et al.*, 2020
Invasive Species Early Detection	Combining high-resolution imagery and AI to identify invasive species and monitor spread.	Enables rapid management interventions, reducing ecological damage	Hyperspectral imaging, machine learning classifiers	Difficult to distinguish invasive species spectrally; ground validation needed	Detection of invasive plant species in wetlands	Shafiq *et al.*, 2024; Lake *et al.*, 2022
Wildlife Corridor and Connectivity Mapping	Mapping landscape connectivity and barriers for key species using spatial and movement data.	Facilitates landscape-scale conservation and connectivity planning	GPS telemetry, landscape resistance modeling	Requires multi-source data fusion; temporal dynamics often underrepresented.	Corridor design for big cats in fragmented habitats	Cisneros-Araujo *et al.*, 2021; Martin *et al.*, 2019
Ecosystem Service Monitoring	Linking biodiversity metrics from RS with ecosystem services such as pollination, carbon, and water.	Informs policy and payment for ecosystem services (PES) schemes	Multispectral indices, spatial modeling, economic valuation models	Complex biophysical-socioeconomic integration; data resolution mismatch	Monitoring carbon stocks in tropical forests	Cimoli *et al.*, 2024; Imran *et al.*, 2021
Real-Time Illegal Activity Surveillance	Using satellite, UAV, and sensor networks to monitor poaching, illegal logging, and land-use changes.	Enhances law enforcement and deters biodiversity crimes.	UAVs, sensor networks, automated change detection	Requires frequent data acquisition and rapid analytics; privacy concerns	Anti-poaching patrols in African savannas	Mangewa *et al.*, 2019; Tripathi *et al.*, 2025
Phenology and Seasonal Monitoring	Tracking seasonal changes in vegetation and species behavior to understand ecosystem dynamics.	Supports adaptive management by revealing phenological shifts.	Time-series analysis of satellite data (e.g., NDVI, EVI)	Cloud cover interference; variable temporal resolution	Monitoring flowering times to assess climate impacts	Gong *et al.*, 2024; Surasinghe *et al.*, 2025
Disease and Pest Outbreak Detection	Identifying early signs of disease or pest infestations affecting biodiversity	Enables timely mitigation to protect vulnerable species and habitats.	Thermal imaging, multispectral sensors, AI pattern recognition	Disease symptoms can be spectrally subtle; requires ground truthing	Locust swarm monitoring using remote sensing	Brown *et al.*, 2022; Kariyanna and Sowjanya, 2024
Marine and Coastal Habitat Mapping	Mapping coral reefs, seagrass beds, and mangroves using underwater and surface remote sensing.	Informs marine spatial planning and coastal ecosystem protection.	Multispectral satellite imagery, sonar, UAV surveys	Water turbidity and depth limit detection accuracy	Coral bleaching monitoring in the Great Barrier Reef	Abo Elenin *et al.*, 2020; Siddiqui and Zaidi, 2024
Biodiversity Hotspot Identification	Using multi-sensor RS to identify regions with high species richness or endemism.	Prioritizes conservation areas and resource allocation.	Multispectral, hyperspectral imaging, biodiversity modeling	Spatial resolution may miss microhabitats; requires field validation	Identifying Amazonian hotspots for protection	Marchese, 2015; Wu and Liang, 2018
Carbon and Biomass Estimation	Estimating forest biomass and carbon stocks as proxies for ecosystem health.	Supports climate mitigation strategies and REDD+ programs.	LiDAR, RAdio Detection and Ranging, multispectral imaging	High cost and processing demands; variable accuracy in dense forests	Estimating carbon stocks in boreal forests	Chinembiri *et al.*, 2024; Ranglong *et al.*, 2025
Landscape Disturbance and Recovery Monitoring	Detecting natural or anthropogenic disturbances and tracking ecosystem recovery over time.	Assesses conservation effectiveness and guides restoration efforts.	Time-series satellite imagery, change detection algorithms	Distinguishing natural from human-induced changes can be complex	Monitoring forest regrowth after fire events	Ren *et al.*, 2023; Tan *et al.*, 2024
Species Distribution and Habitat Suitability Modeling	Predicting spatial distribution of species based on habitat variables derived from RS data.	Supports targeted species conservation and habitat management.	GIS-based habitat models, machine learning, environmental layers	Model uncertainty; lack of fine-scale data for rare species	Habitat suitability for endangered amphibians	Dastres *et al.*, 2025; Rew *et al.*, 2020

Socio-ecological integration and ethical dimensions

Inclusion of indigenous and local knowledge

Participatory geographic information systems (PGIS) have become crucial tools for integrating indigenous and local knowledge into natural resource management and land use planning, especially in contexts such as rangeland management in Africa. Azong Cho and Mutanga (2021), reviewed PGIS applications in African rangeland use planning and highlighted how PGIS facilitates the capture of diverse local actors' perspectives, contributing to more inclusive and culturally relevant decision-making processes. While PGIS offers promising opportunities for the inclusion of community insights, it still faces challenges, including issues related to content quality, methodological rigor, representation, equity, and access to technology. The study highlights that while PGIS can enhance the empowerment of local communities by allowing them to map and convey their resource utilization and priorities, advantages are frequently obstructed by inconsistent participation and ongoing power disparities. The tourism industry also increasingly leverages public participation geographic information systems (PPGIS) to enhance community involvement in planning and decision-making, ensuring that development is economically viable, environmentally responsible, and socially inclusive. Gheitasi, Salari, and Clark (2024), conducted a comprehensive review of PPGIS use in tourism from 2000 to 2024, identifying that PPGIS provides a credible and flexible alternative to traditional public engagement methods such as meetings and interviews. The research indicated that PPGIS allows stakeholders, such as residents and local businesses, to spatially articulate their concerns and preferences, resulting in more informed and community-focused tourism planning. This digital, geospatial approach supports sustainable tourism by capturing nuanced local knowledge and values, facilitating transparent communication, and increasing the inclusiveness of planning processes. Beyond sector-specific applications, the broader academic landscape of Public Participation GIS (PPGIS) research has evolved substantially since the 1990s, as detailed by Mohamed (2025). In Table 3, key features of stakeholder engagement and the integration of social dimensions into biodiversity monitoring and conservation efforts are summarized.

Data sovereignty and governance

Ensuring data sovereignty in open-access conservation platforms involves more than simply controlling where data is stored; it requires clear frameworks that define ownership rights, usage boundaries, and the responsibilities of all actors involved. Without proper governance, there is a risk that sensitive ecological or community information could be misappropriated, which would weaken trust and dissuade involvement from important stakeholders. This is particularly critical in conservation contexts where data often relates to endangered species locations, indigenous territories, or vulnerable habitats. In these situations, misuse could lead to ecological harm or social exploitation. Supporting this, Brodny and Tutak (2025), emphasize that without robust ethical integration in technological systems, even innovations that are well-intentioned may jeopardize privacy and undermine public trust. Their work shows that transparent stakeholder engagement is essential for balancing technological advancement with the protection of fundamental rights. Governance frameworks should also tackle the imbalanced power dynamics present in AI-driven conservation efforts, guaranteeing that advantages are equitably distributed and that marginalized communities are included in the decision-making process. This includes creating equitable access to AI tools, transparent data-sharing agreements, and mechanisms to challenge misuse or bias. In the context of higher education, Balan (2024), emphasizes that sustainable AI integration requires equity-focused policies, proactive bias mitigation, and inclusive governance structures.

Monitoring for environmental justice

Monitoring for environmental justice through RS–GIS tools requires more than simply mapping physical environmental parameters; it demands a spatially explicit approach that integrates socio-economic vulnerability, ecological stressors, and governance priorities to identify inequality hotspots and guide targeted interventions. Furthermore, without these frameworks, the allocation of resources could remain unaware of the layered challenges that marginalized groups encounter, thus sustaining current disparities. Evidence from Tanzania demonstrates by Macharia *et al.* (2020), combined biophysical and socio-economic

Table 3: Stakeholder engagement and social dimensions in biodiversity monitoring

Aspect	Description	Benefits	Challenges	Key Stakeholders	Implementation Examples	Success Factors	Sources
Incorporation of Traditional Ecological Knowledge (TEK)	Integrating indigenous and local ecological knowledge systems with modern monitoring tools.	Provides context-rich insights; improves monitoring relevance; strengthens cultural identity.	Cultural sensitivity; protecting intellectual property rights; potential mistrust.	Indigenous communities, elders, cultural heritage bodies, ecologists.	Arctic biodiversity monitoring using Inuit knowledge; Amazon forest mapping with tribal input.	Trust-building, equitable data-sharing agreements, recognition of intellectual property.	Bwambale *et al.*, 2018; Singleton *et al.*, 2023
Community-Based Monitoring (CBM)	Local communities actively collect and interpret biodiversity data.	Builds local capacity; fosters stewardship; enables rapid detection of changes.	Ensuring data quality; sustaining long-term participation; avoiding volunteer fatigue.	Local NGOs, residents, schools, conservation agencies.	Reef monitoring by coastal villages; forest patrols by community rangers in Africa.	Continuous training, incentives, feedback loops, clear communication of impact.	Chandler *et al.*, 2017; Price *et al.*, 2023; Shephard *et al.*, 2023
Open Data and Citizen Science Platforms	Publicly accessible biodiversity data portals and apps enabling citizen participation.	Expands spatial/temporal coverage; democratizes science; fosters transparency.	Data verification; digital divide; varying skill levels among contributors.	General public, tech developers, universities, conservation agencies.	iNaturalist, eBird, Global Biodiversity Information Facility (GBIF).	Robust data validation, user-friendly platforms, offline data collection options.	Serbe-Kamp *et al.*, 2023; Ziliaskopoulos and Laspidou, 2024
Policy Integration and Adaptive Management	Linking monitoring outputs with governance and adaptive policy frameworks.	Enables responsive conservation; supports evidence-based decision-making.	Bureaucratic inertia; conflicting stakeholder priorities; slow policy cycles.	Government ministries, policy-makers, NGOs, researchers.	European Union Habitats Directive reporting; adaptive wildlife corridor management.	Cross-sector coordination, policy champions, legal mandates for monitoring data use.	Armstrong and Kamieniecki, 2017
Cross-Border Biodiversity Collaboration	Shared monitoring and conservation efforts across political boundaries.	Protects migratory species; improves regional ecosystem resilience.	Political tensions; harmonizing methodologies; data sovereignty issues.	Neighboring governments, transboundary conservation groups, intergovernmental bodies.	Great Limpopo Transfrontier Park wildlife tracking; Mekong River fish monitoring.	Diplomatic agreements, standardized protocols, shared funding.	Kotru *et al.*, 2020; Magory Cohen *et al.*, 2022
Private Sector Engagement	Partnering with industries for biodiversity data sharing and sustainable practices.	Access to new data sources (e.g., corporate landholdings); co-funding opportunities.	Conflicts of interest; greenwashing risks; transparency issues.	Agribusiness, mining companies, tourism operators.	Biodiversity offsets by mining companies; wildlife-friendly farming certifications.	Clear agreements, independent oversight, long-term sustainability commitments.	Ackers and Adebayo, 2024
Education and Awareness Campaigns	Using biodiversity monitoring outcomes to inform and educate.	Builds public support; inspires conservation action.	Over-simplification risk; competing public priorities.	Schools, media outlets, NGOs.	Public campaigns on pollinator declines; youth biodiversity clubs.	Engaging storytelling, culturally relevant content, multi-platform outreach.	Ardoin *et al.*, 2020; Banha *et al.*, 2022
Participatory Mapping and GIS Storytelling	Co-creating maps and spatial narratives with stakeholders.	Encourages local ownership; improves data accuracy with lived experience.	Potential disagreements on territorial boundaries; data sensitivity.	Local communities, GIS experts, advocacy groups.	Participatory wetland mapping in South America; indigenous land rights mapping in Canada.	Inclusive facilitation, accessible map tools, respect for confidentiality.	Onencan *et al.*, 2018; Weyer *et al.*, 2019

indicators to produce climate vulnerability index maps for the Wami-Ruvu and Rufiji basins, revealing concentrated "hotspot" communities where water scarcity, climate variability, and population growth intersect to intensify risk. These spatial outputs were employed directly to prioritize measures for adaptation and to bolster local institutional capacity, illustrating how geospatial analysis can move beyond diagnosis to actionable resilience planning. Similarly, Boori *et al.* (2022), applied a DPSIR-based ecological vulnerability index in Russia's Republic of Tatarstan, integrating 23 indicators from satellite, socio-economic, and demographic data to capture the cumulative pressures of industrialization and environmental degradation.

Data integration and standardization

Harmonizing heterogeneous multi-sensor and multi-resolution remote sensing datasets remains a critical challenge for advancing biodiversity monitoring. Differences in spatial, spectral, and temporal resolutions, as well as varying sensor characteristics and coordinate systems, may cause inconsistencies that obscure ecological signals unless they are meticulously handled. Developing standardized pre-processing workflows, calibration protocols, and cross-sensor harmonization techniques is essential to ensure that the ecological integrity of the data is preserved while enabling meaningful comparisons across studies and ecosystems (Aguzzi *et al.*, 2025). It is equally vital to standardize RS-derived indicators that assess habitat quality and ecosystem function. Scaling locally validated RS/GIS methodologies to global applications presents additional complexity, as models trained in specific ecological contexts may not generalize without introducing oversimplification (Markham *et al.*, 2023).

Biodiversity detection and measurement

Remote sensing has significantly progressed in the mapping of habitats and the evaluation of ecosystem-level patterns; however, identifying cryptic or rare species continues to be a persistent challenge. The inherent spatial and spectral limitations of many sensors mean that individuals or low-density populations can be overlooked, potentially underestimating biodiversity. Integrating remote sensing with complementary approaches, such as environmental deoxyribonucleic acid (eDNA) sampling and bio-logging, can bridge this gap. eDNA provides species-level detection without direct observation, while bio-logging tracks individual movements and behaviors (Li *et al.*, 2024). When these approaches are paired with high-resolution spatial data, they create a more comprehensive and accurate representation of biodiversity distribution, particularly for elusive or nocturnal species. Moving beyond habitat proxies toward direct measurements of functional, trait, and genetic diversity represents a critical frontier in RS applications. Hyperspectral imaging, structural metrics derived from LiDAR, and observations from drones can detect fine differences in leaf chemistry, canopy structure, and species compositions that indicate functional characteristics (Shu *et al.*, 2025). Trait-based indicators can be connected to ecosystem processes, which allows researchers to derive biodiversity function rather than relying only on surrogate habitat metrics. Incorporating genetic or phylogenetic information, either through eDNA or trait-environment correlations, further enhances the ecological interpretability of remotely sensed data, providing a more mechanistic understanding of ecosystem dynamics (Bauknecht *et al.*, 2025). Identifying invasive species at an early stage is yet another crucial application in which next-generation RS exhibits promise. Hyperspectral sensors, coupled with drone platforms, can identify subtle spectral anomalies or morphological signatures indicative of emerging invasive populations (Karahan *et al.*, 2025). Integrating these early-warning detections with prompt response measures, including targeted management interventions or alert systems, can notably decrease ecological and economic impacts. Such integration requires automated change detection, real-time processing, and seamless communication between monitoring and management frameworks. The latest remote sensing technologies also provide the potential to advance fine-scale models of species distribution and habitat suitability. By integrating multi-temporal and multi-resolution datasets with ecological niche modeling, researchers can dynamically update predictions for both endangered and invasive species in response to environmental change (Devika and Amitha Bachan, 2025).

Forecasting and early-warning systems

Remote sensing and GIS are progressively being

seen not merely as tools for observing biodiversity, but also as systems for anticipating ecological transformations under various climate and land-use conditions. By integrating multi-temporal and multi-resolution datasets with predictive modeling frameworks, RS/GIS can anticipate shifts in species distributions, habitat quality, and ecosystem functioning, enabling proactive management interventions (Bin *et al.*, 2025). Near-real-time data streams assist in the development of early-warning systems that can detect emerging threats, ecological tipping points, or potential regime shifts ahead of irreversible damage. Time-series analyses of satellite and drone-derived observations can be optimized to disentangle climate-driven variability from anthropogenic pressures, improving the reliability of forecasts (Inoue, 2020). RS-derived phenological indicators, such as leaf-out timing, flowering, or migration cues, can be linked to species interactions to identify potential ecological mismatches under changing climates, while coupled vegetation; climate models allow for the prediction of feedback loops that influence ecosystem resilience (Renner and Zohner, 2018). Employing GIS to provide spatial context to these dynamics facilitates the detection and modeling of vital recovery thresholds, tipping points, and regime shifts in complex ecosystems (Dakos *et al.*, 2024).

Ecosystem functions, networks, and services

Remote sensing and GIS provide powerful tools for moving beyond static habitat mapping to capture the dynamic integrity of ecological networks, including habitat connectivity, trophic interactions, and pollination systems. By merging multi-source data including LiDAR, radar, and optical imagery, RS techniques are capable of quantifying structural and functional connectivity, observing changes in wildlife corridors over time, and validating these trends with genetic flow data to ensure ecological authenticity (Neumann *et al.*, 2015). Linking biodiversity metrics with ecosystem service assessments enables spatially explicit evaluation of provisioning, regulating, and supporting services, informing conservation prioritization and landscape management decisions (Ramel *et al.*, 2020). By merging LiDAR, radar, and optical datasets facilitates accurate estimations of biomass and carbon stocks both above and below ground, effectively aligning biodiversity monitoring with climate mitigation efforts (Luo *et al.*, 2017). Advanced time-series analyses and change-detection algorithms can identify subtle or cumulative disturbances, monitor successional pathways, and differentiate between natural variability and anthropogenic influences, aiding inrestoration and recovery efforts. Furthermore, deep-learning-enabled analysis of high-resolution RS imagery has the potential to uncover previously unknown biodiversity hotspots or ecological corridors that are not captured by field-based surveys, thereby expanding the scope and effectiveness of conservation planning (Pettorelli *et al.*, 2025).

Analytical and computational innovations

Next-generation remote sensing technologies, including hyperspectral sensors, LiDAR, radar, small satellite constellations, and drones, are dramatically enhancing biodiversity and ecosystem monitoring capabilities by providing high-resolution, multi-dimensional, and temporally continuous data. These sensors provide the capability for detailed mapping of species distributions, habitat structures, and ecosystem functions at extraordinary scales, which helps in recognizing subtle ecological variations and patterns that have not been previously documented. Machine learning and deep learning models offer powerful analytical frameworks for processing these complex datasets, but ensuring transparency, interpretability, and freedom from algorithmic bias is essential for their reliable application in conservation (Patil *et al.*, 2024). Implementing explainable AI strategies, assessing uncertainty, and ensuring rigorous model validation are crucial for building confidence among practitioners and policymakers (Mersha *et al.*, 2024).

Governance, access, and ethics

Remote sensing and GIS-based biodiversity monitoring systems offer unprecedented opportunities to empower Indigenous and local communities while safeguarding cultural sovereignty and intellectual property rights. Collaborating with these stakeholders to co-develop monitoring frameworks allows data collection and interpretation to align with local knowledge systems and priorities, thereby promoting ownership, trust, and equitable benefit-sharing (Ćwikła *et al.*, 2025). Real-time RS-based surveillance can enhance conservation

enforcement by detecting illegal activities such as logging, mining, or poaching, yet these systems must be designed to respect privacy and sovereignty, ensuring that monitoring does not inadvertently infringe upon community rights (Wang *et al.*, 2025). Additionally, RS/GIS workflows present strong and consistent indicators that can facilitate global biodiversity accounting frameworks and national reporting obligations, including the Sustainable Development Goals and the Kunming–Montreal Global Biodiversity Framework.

Challenges and knowledge gaps

One of the most urgent challenges in applying RS and machine learning (ML) to biodiversity monitoring is the limited availability of ground-truthed biodiversity indicators for algorithm training. Furthermore, lacking comprehensive in situ datasets can cause predictive models to misidentify habitats or miss subtle ecological shifts, potentially leading to erroneous conservation choices. Field validation remains essential for refining models and ensuring that automated classifications accurately reflect ecological conditions on the ground. For instance, Kovárník and Janová (2025) demonstrated that validating Sentinel-2-based ML models with National Forest Inventory data significantly improves model accuracy, achieving up to 98.3% in land cover estimation. A further important issue is the undue reliance on proxy metrics, including spectral indices, instead of relying on direct ecological observations. Furthermore, while NDVI can indicate broad patterns, it can also overlook the complexities of ecological variability in complex environments. Pinon *et al.* (2024) highlighted that although spectral diversity correlates strongly with species richness and diversity in the Brazilian Atlantic Forest, factors such as pixel size and shadow effects influence these relationships, underscoring the need to integrate spectral proxies with field-based observations for a more comprehensive understanding of biodiversity dynamics. The significant computational requirements for handling multi-source RS data pose a short-term challenge, which is becoming more manageable with contemporary workflows. Integrating hyperspectral, LiDAR, radar, and multispectral datasets requires cloud-based platforms and specialized algorithms capable of handling petabyte-scale inputs. Prošek *et al.* (2020) showed that combining hyperspectral and LiDAR data significantly improves classification accuracy, particularly for small water bodies, with omission and commission errors below 10%. These findings highlight that adopting advanced computational workflows can mitigate processing challenges in the near term. Likewise, challenges related to interoperability differing sensor formats, spatial resolutions, and coordinate systems make it difficult to achieve seamless integration of datasets sourced from multiple origins. Differences in temporal coverage, geodetic reference frames, and preprocessing requirements increase operational costs and the risk of analytical errors. In conclusion, limitations in satellite sensor capabilities present an ongoing challenge for biodiversity observation. Often, current sensors are deficient in the integrated spatial, spectral, radiometric, and temporal resolutions required to accurately capture swiftly changing ecological variables, particularly in dynamic coastal environments. Muller-Karger *et al.* (2018) emphasize the need for a new generation of satellites with enhanced specifications, referred to as H4 imaging, to improve data consistency and interoperability.

Future frontiers and research directions

The integration of eco-acoustics and bio-logging data into remote sensing and GIS platforms offers a significant advancement in biodiversity monitoring. By combining environmental audio recordings with spatial data on species movement, researchers can obtain a richer, multidimensional perspective of ecosystems. This technique enhances the ability to identify elusive or nocturnal species that are frequently overlooked by standard remote sensing methods, improving both the range and accuracy of biodiversity evaluations. Farina (2018) notes that ecoacoustics provides quantitative metrics that serve as proxies for biodiversity and ecosystem health, supporting both field-based and remote surveys while enabling long-term monitoring across multiple spatial and temporal scales. Combining these datasets with information obtained from satellites can improve predictive modeling, especially in regions where data is scarce, by identifying ecological variability that purely spectral metrics could fail to recognize. Developing near-real-time biodiversity observatories using edge artificial intelligence and in-situ sensors represents another frontier. Processing data locally at sensor sites reduces latency and bandwidth requirements,

enabling the rapid detection of emerging threats such as invasive species, habitat degradation, or sudden population declines. This forward-thinking monitoring method can boost conservation responsiveness and adaptive management. Ditria *et al.* (2022) highlight that AI-driven automated monitoring addresses key challenges in conservation, including limited funding, workforce constraints, and bottlenecks in data processing, thereby facilitating timely access to comprehensive information for informed decision-making. Kovari (2024) points out that achieving a balance between transparency, accuracy, and trust is essential for the successful implementation of AI-based decision support systems across different industries. Explainable AI enables stakeholders to assess uncertainties, evaluate risks, and adopt interventions with greater confidence, ultimately improving the effectiveness of technology-guided conservation strategies. Some proposed innovations extend beyond RS–GIS and AI to cross-disciplinary approaches. The combination of biodiversity monitoring and economic models, including biodiversity credits, is an innovative approach that connects ecological results to financial rewards. Integrating spatial biodiversity data with market-based mechanisms can help prioritize conservation investments toward interventions with the highest ecological impact. Wunder *et al.* (2025) report that biodiversity credits offer potential for enhancing pro-environmental behavior, although challenges remain in their design, governance, and impact evaluation. Co-designing monitoring systems with local stakeholders, including communities, indigenous groups, and land managers, is vital for long-term sustainability and relevance.

Future research priorities

- Develop integrated RS–GIS platforms that combine eco-acoustic, bio-logging, and satellite data to enable multidimensional and high-resolution biodiversity monitoring;
- Develop advanced AI technologies and automated, near-real-time observatories to facilitate improved early detection of ecological risks and alleviate data processing bottlenecks;
- Enhance AI explainability and transparency to ensure practitioner trust, informed decision-making, and ethical deployment in conservation;
- Explore cross-disciplinary approaches, such as biodiversity credits, ensuring robust monitoring and governance frameworks to link ecological outcomes with policy and economic incentives;
- Engage local stakeholders through participatory co-design methods to improve the relevance, accessibility, and sustainability of monitoring systems over time.

CONCLUSION

The integration of remote sensing and geographic information systems has become increasingly indispensable for biodiversity monitoring and ecosystem management under accelerating climate change, land-use transformations, and other anthropogenic pressures. This review specifically underscores the unique contributions of emerging RS–GIS technologies to the discipline, transcending a basic summary of tools to illustrate how these innovations are altering ecological science and conservation practice. By combining multi-sensor data fusion, advanced machine learning and deep learning frameworks, and participatory monitoring approaches, RS–GIS applications now enable the generation of high-resolution, temporally continuous, and spatially explicit datasets. These datasets deliver exceptional understanding of biodiversity dynamics, functional characteristics, and ecosystem processes at various scales; local, regional, and global; equipping conservation practitioners and policymakers with actionable intelligence. Recent advancements, including hyperspectral imaging, drone-based sensing, radar interferometry, and CubeSat constellations, have significantly enhanced the ability to detect fine-scale ecological patterns, species distributions, and habitat changes in near real-time. When coupled with analytical tools such as convolutional neural networks, transfer learning, and cloud-based geospatial platforms, these technologies facilitate automated classification, ecosystem change forecasting, and detection of subtle phenological and landscape-level shifts that were previously undetectable. Moreover, the incorporation of multi-temporal and multi-resolution datasets improves the ability to monitor ecological processes, evaluate connectivity, and observe environmental changes with greater accuracy, thereby guiding focused management strategies. Despite these transformative capabilities, persistent challenges limit the full operationalization of RS–GIS in biodiversity science. The scarcity of ground-truthed

biodiversity information hampers the calibration and validation of models, which in turn creates in ecological predictions. Interoperability issues among heterogeneous sensors, platforms, and analytical frameworks complicate data integration, while reliance on proxy variables such as vegetation indices can reduce ecological interpretability. High computational requirements and the unavailability of accessible high-performance infrastructure continue to hinder large-scale analyses, particularly in regions with low and middle incomes. Beyond technical considerations, ethical, social, and governance challenges, including data sovereignty, equitable benefit-sharing, and inclusion of indigenous and community knowledge, remain critical for ensuring socially just and globally relevant monitoring systems. This review underscores several actionable priorities for advancing RS–GIS applications in biodiversity science. Integrative frameworks that combine remote sensing technologies with in-situ ecological observations, citizen science contributions, contributions from citizen science, and engagement with local stakeholders are crucial for enhancing the accuracy and relevance of models. Expanded field-based validation campaigns and open-access, interoperable platforms will democratize the use of RS–GIS tools, enabling broader participation in ecological monitoring. Developing effective data governance systems that harmonize transparency, security, and inclusivity is crucial for the long-term application of these technologies. By addressing both technical and socio-ecological challenges, RS–GIS can become a foundational component of global biodiversity intelligence, supporting evidence-based conservation strategies, adaptive management, and proactive policy formulation. The use of RS–GIS applications will enhance the recording of ongoing biodiversity changes while also delivering actionable insights that can reinforce ecosystem resilience, protect essential habitats, and ensure the long-term sustainability of life-support systems on which human well-being depends. This review contributes to the field by outlining practical pathways for integrating technological, ecological, and societal dimensions, enabling the development of predictive, equitable, and impactful biodiversity monitoring systems.

AUTHOR CONTRIBUTIONS

A. Zhidebayeva conceptualized the review framework, coordinated the overall structure, and contributed to drafting and editing the manuscript. S. Syrlybekkyzy conducted an extensive literature search, performed data extraction, and contributed to synthesizing key findings. L. Taizhanova organized and analyzed the reviewed literature, prepared summary tables, and assisted in drafting the methodology and results synthesis. S. Koibakova contributed to the selection and critical evaluation of relevant studies and assisted in preparing figures and visualizations. Z. Altybaeva supported the literature review process, compiled supporting references, and assisted in drafting the discussion section. A. Koishina participated in structuring the manuscript, reviewed emerging trends, and assisted in finalizing the conclusions. L. Seidaliyeva contributed to summarizing recent advancements, refined the narrative flow, and critically reviewed the manuscript draft. T. Mkilima, the corresponding author, conceived the review idea, supervised the entire process, integrated contributions from all authors, critically revised the manuscript, and finalized it for submission.

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CONFLICT OF INTEREST

The authors declare no potential conflict of interest regarding the publication of this work. In addition, the ethical issues including plagiarism, informed consent, misconduct, data fabrication and, or falsification, double publication and, or submission, and redundancy have been completely witnessed by the authors.

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Negative AI Statement

The author(s) declare that no AI tools or services were not used or not highly applied during the preparation of this work.

ABBREVIATIONS

%	Per cent
AI	Artificial intelligence
BGI	Blue-green infrastructure
CICDDOS 2019	Canadian Institute for Cybersecurity Distributed Denial of Service 2019 Dataset
CIgreen	Green chlorophyll index
CIred edge	Red-edge chlorophyll index
CNNs	Convolutional Neural Networks
CO_2	Carbon dioxide
CubeSat	Cube Satellite
CWC	Canopy water content
DL	Deep learning
eDNA	Environmental deoxyribonucleic acid
e.g.	Exempli gratia/ for example
EnMAP	Environmental Mapping and Analysis Program
ESIS	Earth sensing imaging spectrometer
EVI	Enhanced vegetation index
EWT	Equivalent water thickness
Fig.	Figure
GEDI	Global ecosystem dynamics investigation
GIS	Geographical information system
GVMI	Global vegetation moisture index
IDI	Integrated disturbance index
LFMC	Live fuel moisture content
LiDAR	Light detection and ranging
LSWI	Land surface water index
ML	Machine learning
MSI	Moisture stress index
NDII	Normalized difference infrared index
NDVI	Normalized difference vegetation index
NDWI	Normalized difference water index
NIR	Near-infrared
NRred edge	Red edge normalized ratio
PGIS	Participatory geographic information systems
PRISMA	PRecursore IperSpettrale della Missione Applicativa
PRI	Photochemical reflectance index
RNN	Recurrent neural network
RS	Remote sensing
SAR	Synthetic aperture radar
SAVI	Soil adjusted vegetation index
SUHI	Surface urban heat island
SWIR	Shortwave infrared
SVM	Support vector machines
T/ha	Ton per hectare
UAVs	Unmanned aerial vehicles
XAI	Explainable AI

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